Sprint 3 Contribution Log - Kaila

Overview

In Sprint 3, I completed my database backend tasks: KA-9 (one login helper), KA-6 (notifications table, interview date column and migration), KA-7 (notification service, with the US-5 acceptance test drafts), KA-8 (demo data script) and KA-10 (requirements and README cleanup), KA-12 (reading text from PDF and Word resumes for the AI feedback feature), and the first part of KA-13 (limiting AI feedback requests). Every task has its own branch, tests and pull request, the migration and the demo data are live on Neon, and I added 50 automated tests (17 for KA-9, 5 for KA-6, 15 for KA-7, 9 for KA-12 and 4 for KA-13).

KA-9 - One Login Helper

Goal: Merge the two login helpers into one, so every protected route answers login problems the same way.

What I did:

Investigation: found that deps.py was used by the profile, resumes, applications and application status routers, and dependencies.py by the jobs and recruiter routers. No test or frontend file checked the error message, and the missing-header case behaved differently, so I used path C then A.

One helper: App/core/dependencies.py now uses HTTPBearer(auto_error=False) and answers 401 "Not logged in." for no header, a non-Bearer header, a bad or expired token, or a deleted account. require_recruiter is unchanged (403 "Recruiter access required").

Imports: pointed the four routers at App.core.dependencies and deleted App/core/deps.py.

Tests: tests/test_auth_helpers.py with 17 tests.

KA-6 - Notifications Table, Interview Date and Migration

Goal: Let the database store notifications and an interview date on each application, so the notification tasks (KA-7, KA-8, LA-7, LA-8, YO-7) can start.

What I did:

Notification model: SQLAlchemy model for notifications with user, type, message, application, read flag, creation time and event key (App/models/notification.py).

Data rules: deleting a user deletes their notifications, and deleting an application keeps the notification but clears the link. A unique rule on user and event key (uq_notification_user_event) stops the same reminder from being created twice.

Interview date: added interview_at to applications (App/models/application.py).

Model registration: imported the model so Alembic detects it (App/models/__init__.py).

Migration: wrote a migration that builds on 16d4c4e691bf and only adds the new table, its index and the new column (alembic/versions/1dae1f3df735_add_notifications_and_interview_at.py).

Tests: tests/test_notification_model.py with 5 tests.

Restore after a revert: KA-6 was merged (#123) and then reverted (#128), which left main without the migration that Neon is on. I brought it back by reverting the revert on a new branch, so exactly the same five changes return (#130).

KA-7 - Notification Service

Goal: Database functions for notifications, so the notification endpoints and the status change only need to call them.

What I did:

Service functions: create_notification, notify_new_application, notify_status_change, list_notifications, count_unread, mark_read and mark_all_read, with all the message wording in one file (App/crud/notifications.py).

No commit inside: the functions add and flush but never commit, so a notification is saved together with the change that caused it.

Ownership: mark_read returns nothing if the notification belongs to someone else.

Interview reminders (P2): generate_interview_reminders creates one reminder for each interview in the next 24 hours, with the interview time in the event key so a rescheduled interview reminds again. No background timer is needed.

Tests: tests/test_notifications_service.py with 15 tests.

Into main: the first pull request (#124) targeted my KA-6 branch and was merged there, so it never reached main. I merged main and the restored KA-6 into the KA-7 branch and opened it straight into main (#131).

Acceptance test (AT-5, #161): wrote the US-5 scenarios in Given / When / Then form: a status change notifies the job seeker, a new application notifies the recruiter, and an interview within 24 hours creates a reminder. Each one points to the service tests that cover it.

KA-8 - Demo Data Script

Goal: One command that fills the database with demo data, so the dashboard, the notifications and the recruiter screens have something to show in the demo.

What I did:

Demo script: a demo recruiter (Aurora Robotics) and a demo job seeker with hashed passwords, 12 jobs (5 posted by the demo recruiter), and 12 applications over the last 30 days covering all five statuses, one of them Interview with the interview tomorrow (scripts/seed_demo_data.py).

Notifications: created with my KA-7 functions: new applications for the recruiter, status changes, an interview invitation and an interview reminder for the job seeker.

Safety: shows the database host and only continues if you type YES, does nothing if the demo recruiter already exists, and commits once at the end. It never runs inside the tests.

Into main: the pull request (#125) first targeted my KA-7 branch. Once KA-7 was in main, I pointed it at main so its CI checks ran, and it was merged.

KA-10 - Requirements and README Cleanup

Goal: requirements.txt has one line per package, and the README says only what really exists.

What I did:

requirements.txt: removed the duplicate pytest>=8.0.0 (kept pytest==9.1.1). httpx and httpx2 were both listed; nothing imports either directly, but the test client now uses httpx2 and warns when it gets httpx, so I kept httpx2 and removed httpx.

Features: the README listed things we don't have (AI feedback, CV matching, favourites). It now lists what exists, marks the Sprint 3 work as in progress, and puts the rest under Planned for Sprint 4.

API and tests: added an API section (all paths start with /api/v1, wrong input returns 400, docs at /docs) and updated Running Tests with the backend and frontend commands, the Windows command, a link to Documentation/TESTING.md and the two seed scripts.

KA-12 - Read Text from PDF and Word Resumes

Goal: One function that turns a resume file into text for the AI feedback feature, with clear errors when there is no text to read.

What I did:

Libraries: added pypdf (PDFs), python-docx (.docx) and fpdf2 (tests only, to build sample PDFs) to requirements.txt with exact versions, in a small separate pull request so the whole team can use them right away (#132).

Text reader: extract_resume_text(content_type, data) in App/services/resume_text.py. PDFs give the text of every page; .docx files give every paragraph and every table cell. Old .doc and other types raise ResumeTextError ("Old .doc files are not supported. Please upload a PDF or .docx file."). The text is cleaned of repeated spaces and blank lines, cut at 12,000 characters, and fewer than 50 characters raises "No readable text was found. A scanned file has no text." Broken or password-protected files raise ResumeTextError, and no error message ever contains resume text.

New folder: App/services/, laid out like App/crud/.

Sample files for tests: tests/resume_samples.py with SAMPLE_RESUME_TEXT, make_pdf_bytes and make_docx_bytes, shared with Lamees and Youssef for their tests.

Tests: tests/test_resume_text.py with 9 tests (#133).

Conflict fix: my updated log PDFs had been committed on the KA-12 branch by mistake, which conflicted with main once my logs pull request (#127) was merged. I merged main into the branch and kept main's PDFs, so #133 only changes the four KA-12 files, and linked it to issue #163.

KA-13 - Limit AI Requests per User per Hour

Goal: Nobody can run up the AI bill: each user gets at most 5 feedback requests per hour.

What I did:

Rate limiter: RateLimiter in App/core/rate_limit.py keeps each user's recent request times in memory. allow(user_id, now=None) records the request and returns True, or returns False after 5 requests in the last 60 minutes; reset() clears everything for tests. feedback_limiter is the shared instance.

Tests: tests/test_rate_limit.py with 4 tests (5 pass and the 6th is refused, passes again after 60 minutes, two users don't share a count, reset clears everyone).

Endpoint design: the feedback endpoint checks feedback_limiter.allow(current_user.id) before doing any work and answers 429 "You asked for feedback several times. Please try again later." when the limit is reached (#173, closes #168).

KA-11 - Personal Sprint 3 Report

Goal: One PDF that shows my Sprint 3 work for individual grading.

What I did: wrote AI_Doc/Sprint 3 Report.pdf with a summary, a table of my tasks with their issues and pull requests, what I did, time spent, what I can explain in the lab, and my AI evidence (Tasks 22 to 38 of my AI usage log). Handed in together with this log and my AI usage log in one pull request into main.

Verification

All tasks are tested and verified: every pull request passes the full test suite, the KA-6 migration is applied to Neon with alembic check reporting no differences, the demo data is loaded on Neon, and requirements.txt installs cleanly in a brand-new environment.

KA-9: grep shows nothing imports deps.py anymore. 124 tests passed (107 before plus 17 new), and the app started with uvicorn with no import errors.

KA-6: 112 tests passed. alembic heads showed one head. I told the team, previewed the SQL with alembic upgrade head --sql (only the notifications table, its index and interview_at), and applied it with alembic upgrade head. alembic current now shows 1dae1f3df735 and alembic check reports no new upgrade operations.

KA-7: 127 tests passed, including: the same event key twice gives one row, a rollback removes an unsaved notification, mark_read refuses another user's notification, and a reminder is created once for an interview 5 hours away and none for one 3 days away or a status other than Interview.

KA-8: typing "yes" in lowercase cancelled without changes. With YES, the first run added 2 accounts, 12 jobs, 12 applications, 9 notifications for the job seeker and 5 for the recruiter. The second run printed "Demo data already exists. Nothing was added." All 127 tests still pass.

KA-6 restore: 129 tests passed on top of the current main, and alembic current shows 1dae1f3df735 (head) again.

KA-7 into main: after merging main and the restored KA-6, 144 tests passed and all three files (the model, the migration and the service) are on the branch.

KA-12: installing the three libraries kept all 124 tests on main passing. With the text reader, 133 tests pass: a PDF's and a .docx's words (including table cells) come back, spaces and blank lines are cleaned, 20,000 characters are cut to 12,000, a PDF with no text, random bytes, .doc and unknown types all raise ResumeTextError, and the error never contains resume text.

KA-12 after the conflict fix: git diff against main lists only the four KA-12 files, and 153 tests pass.

KA-13: 144 tests on main before, 148 after adding the limiter and its 4 tests.

KA-10: installed requirements.txt in a brand-new environment and ran the tests: 107 passed with httpx, but with a deprecation warning; with httpx2 instead, 107 passed with no warning.

Work Log

Nov 14, 2026 - Report and logs: Wrote my personal Sprint 3 report (KA-11) and updated this log and my AI usage log. Checked that KA-12 (#133) still merges cleanly into main. Hours: about 1.

Nov 13, 2026 - LA-7 merge: Merged Lamees's notification endpoints (#189), which use my KA-7 functions: GET /notifications creates interview reminders first, and every endpoint commits after calling them. Hours: about 0.25.

Nov 12, 2026 - Labels: Installed the GitHub CLI on my second computer and added the sprint, priority and area labels to all my Sprint 3 pull requests (#122 to #173) and my older Sprint 1 and Sprint 2 pull requests. Hours: about 0.5.

Nov 8, 2026 - KA-8, KA-12, KA-13 and issues: Pointed KA-8 (#125) at main after KA-7 was merged, and deleted the merged KA-6, KA-7 and KA-9 branches so nothing could target them again. Fixed the conflict on KA-12 (#133) by taking my log PDFs out of it. Closed my finished issues (KA-6 #155, KA-7 #156, KA-8 #160) and linked KA-12 to #163. Wrote the KA-13 rate limiter and its tests and opened its pull request (#173). Hours: about 1.25.

Nov 7, 2026 - KA-12: Read the new AI resume feedback tasks. Added pypdf, python-docx and fpdf2 to requirements.txt and opened PR #132. Wrote the resume text reader, the sample-file builders and 9 tests, and opened PR #133. Hours: about 1 (libraries 0.25, text reader and tests 0.75).

Nov 5, 2026 - KA-6 and KA-7 into main: Found that KA-6 had been reverted (#128) and KA-7 (#124) had been merged into the old KA-6 branch, so neither was in main. Restored KA-6 by reverting the revert (PR #130), merged it and main into the KA-7 branch, and opened KA-7 straight into main (PR #131). Explained what happened to Sayf. Hours: about 0.75.

Nov 4, 2026 - Setup, KA-9, KA-6, KA-7, KA-8, KA-10: Set up the project on a second computer (venv, requirements, Git identity, .env). Merged the two login helpers (KA-9) and opened PR #122. Built the notifications table and interview_at column, applied the migration to Neon, and opened PR #123. Wrote and tested the notification service (KA-7), opened PR #124, posted the function names for Lamees and Youssef, and wrote the US-5 acceptance test drafts. Wrote the demo data script (KA-8), loaded the demo data on Neon, and opened PR #125. Cleaned requirements.txt and the README (KA-10), checked them in a clean environment, and opened PR #126. Added my contribution and AI logs (PR #127). Hours: about 3.25 (setup 0.25, KA-9 0.75, KA-6 0.75, KA-7 0.5, KA-8 0.5, KA-10 0.25, acceptance test drafts 0.25).